Green Roof Implementation for Flood Mitigation

A Case of Kissena Neighbourhood, Queens Borough

Project Process Documentation

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Course: LA 5580 — Web Mapping and Data Visualization

1. Project Overview

This document presents the complete step-by-step methodology used to conduct a GIS-based suitability analysis for green roof implementation in the Kissena Neighbourhood of Queens Borough, New York City. The project aims to identify buildings that are best positioned for green roof installation as a nature-based solution for urban flood mitigation.

Buildings were evaluated across six weighted criteria — rooftop area, building age, sunlight availability, building use, water availability, and building height — each reclassified on a standardized 1–5 suitability scale and combined into a composite suitability score using weighted overlay analysis in ArcGIS Pro.

2. Overall Workflow

The figure below summarises the end-to-end process, from criteria selection through to final publication of the interactive dashboard on GitHub Pages.

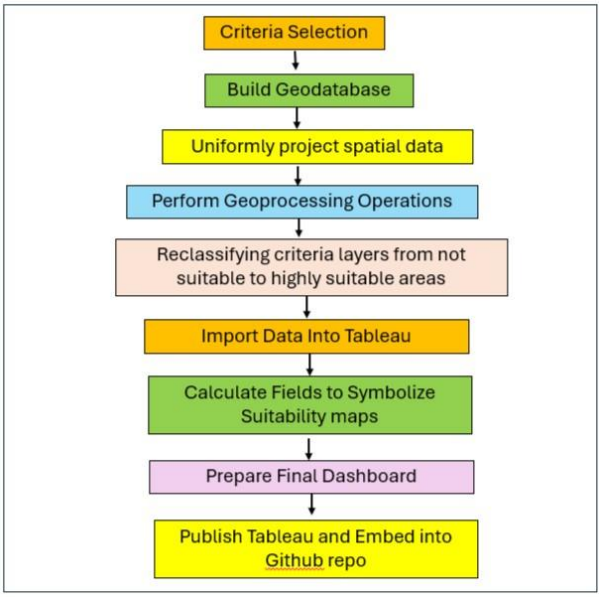

Figure 1. Project workflow diagram — from criteria selection to Tableau publication and GitHub deployment.

3. Step 1 — Criteria Selection and Suitability Classification

The analysis began with a systematic review of green roof suitability literature to identify the most relevant building and environmental characteristics. Six criteria were selected, each assigned a

percentage weight that reflects its relative importance to green roof performance and flood mitigation potential.

Rooftop area sq.m- 20% Weight				
More than 2,154	1,616-2,153	1,077-1,615	401-1,076	Less than 400
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable
Building Age (Year)- 15% Weight				
Less than 20	21-30	31-40	41-80	More than 80
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable
Sunlight Availability kWh/m2/year)- 20% Weight				
More than 2560	2160–2560	1760–2160	1360–1760	Less than 1360
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable
Building Use- 13% Weight				
Public	Residential	Mixed	Commercial	Industrial
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable
Water Availability- 20% Weight				
Highly Available around the year	Available Around the year	Scarcity of water for a short time	Scarcity of water for a long time	Scarcity of water
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable
Building Height (m)- 12% Weight				
4	8	12	16	20
Highly Suitable	Suitable	Moderately Suitable	Less Suitable	Not Suitable

Figure 2. Criteria classification table showing suitability thresholds and percentage weights for each of the six evaluation factors.

The criteria and their weights are summarised below:

- Rooftop Area (sq.m) — 20% Weight: Larger rooftops can support more extensive green roof installations with greater stormwater retention capacity.
- Sunlight Availability (kWh/m²/year) — 20% Weight: Solar radiation drives plant growth and evapotranspiration; higher sunlight supports healthier vegetation layers.
- Water Availability — 20% Weight: Proximity to groundwater and water infrastructure ensures irrigation and system performance year-round.
- Building Age (Year) — 15% Weight: Newer buildings are more likely to meet structural load requirements for green roof systems.
- Building Use — 13% Weight: Public and residential buildings offer greater community benefit; industrial uses are less compatible.
- Building Height (m) — 12% Weight: Lower buildings are easier to access for installation and maintenance, and experience less wind loading.

All criteria were reclassified on a uniform 1–5 scale: 1–2 indicating no suitability to less suitable, 3 indicating moderately suitable, and 4–5 indicating suitable to highly suitable. This common scale enables raster calculator addition and weighted overlay analysis across all layers.

4. Step 2 — Building Geodatabase and Shapefile Assembly

With the criteria defined, the next step was assembling the spatial datasets needed for analysis. All shapefiles were gathered, projected into a uniform coordinate system, and organized within an ArcGIS geodatabase.

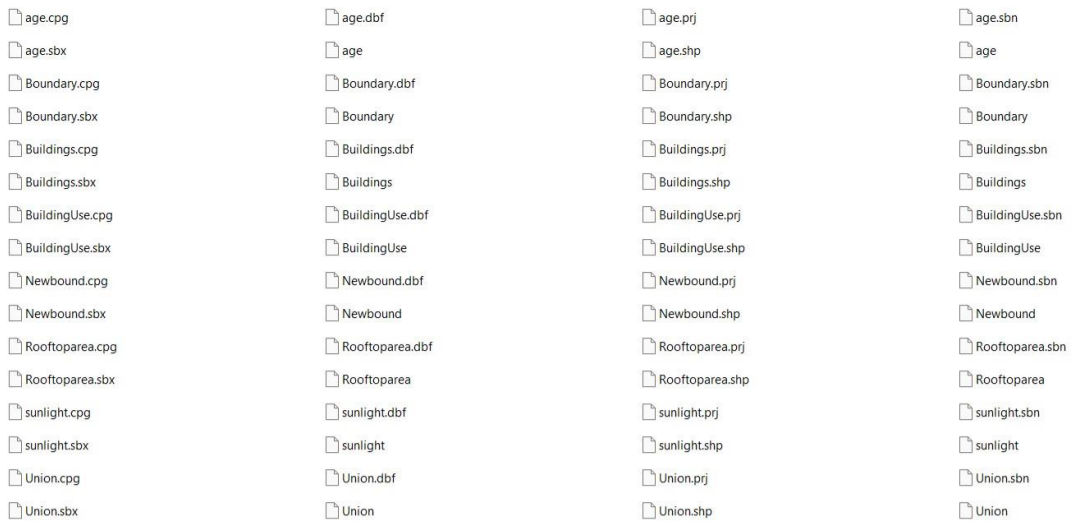

Figure 3. Shapefiles assembled in the project geodatabase — including building footprints, boundary, building use, rooftop area, sunlight, age, and union layers.

The following spatial datasets were incorporated into the geodatabase:

- Building Footprints — sourced from NYC Department of City Planning, providing the spatial extent of every building within the study area.
- Kissena Neighbourhood Boundary — used to clip all datasets to the study extent.
- Building Use (BuildingUse.shp) — NYC Open Data classification of occupancy type per building.
- Building Age (age.shp) — year of construction for each building footprint.
- Rooftop Area (Rooftoparea.shp) — calculated rooftop surface area in square metres.
- Sunlight (sunlight.shp) — solar radiation output derived from DSM processing.
- Union.shp — a merged layer combining attributes from multiple sources for the final weighted overlay.

- DSM Elevation Data — obtained from USGS for solar radiation modelling; clipped to the Kissena boundary before processing.

Spatial joins were performed to attach building-level attribute data (DSM/solar, age, use, rooftop area) to each building footprint polygon. The Solar Radiation tool in ArcGIS was run on the clipped DSM to generate a solar availability raster, which was then spatially joined back to the building footprint layer.

5. Step 3 — Geoprocessing and Raster Analysis

With all layers assembled and joined, geoprocessing operations were performed to reclassify each criterion into the standardised 1–5 suitability scale and produce individual suitability maps per criterion. The Raster Calculator tool was then used to add all reclassified layers together, producing the composite suitability surface. Weighted Overlay Analysis was subsequently applied to incorporate the percentage weights assigned to each criterion, ensuring that the final suitability map correctly reflects the relative importance of each factor.

The weighted overlay weights applied were: Rooftop Area 20%, Sunlight Availability 20%, Water Availability 20%, Building Age 15%, Building Use 13%, and Building Height 12%.

6. Step 4 — Data Import and Calculated Fields in Tableau

Once the suitability analysis was complete in ArcGIS Pro, the processed shapefiles were exported and imported into Tableau Desktop for interactive visualisation. Calculated fields were created in Tableau to translate the numerical gridcode values (1–5) produced by the reclassification process into descriptive suitability labels, enabling intuitive map symbolisation.

6.1 Creating Calculated Fields

For each criterion layer, a calculated field was defined using the following IF-ELSEIF logic pattern, mapping the gridcode integer to suitability text:

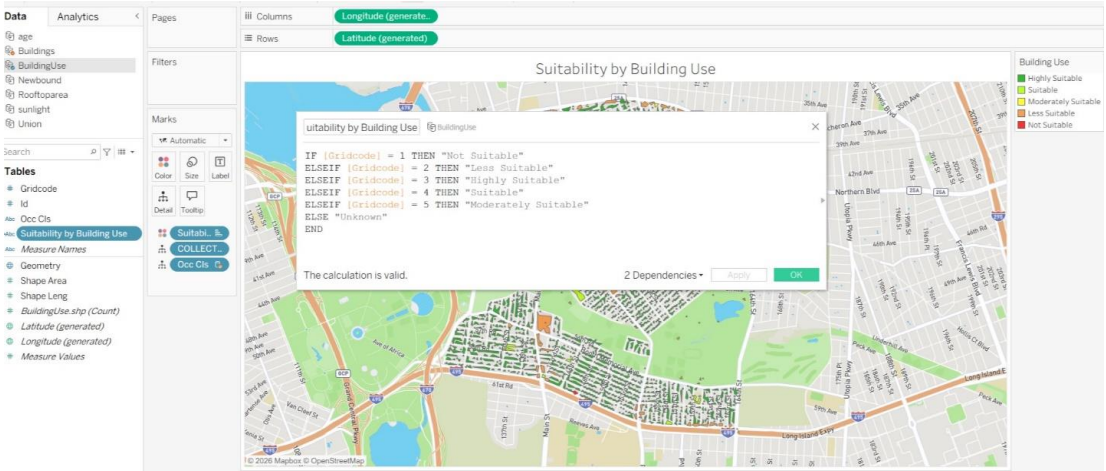

Figure 4. Tableau calculated field for "Suitability by Building Use" — mapping gridcode values to descriptive suitability categories (Not Suitable through Highly Suitable).

The same calculated field structure was applied to all six criterion layers, enabling consistent suitability labelling across all map sheets in the dashboard.

6.2 Configuring Background Map Layers

Background map settings were configured in Tableau to display the Kissena neighbourhood in geographic context. The Outdoors map style was selected, with Streets, Highways, and Routes enabled to provide spatial reference. This background helps stakeholders and viewers understand the geographical distribution of suitable buildings relative to roads and infrastructure.

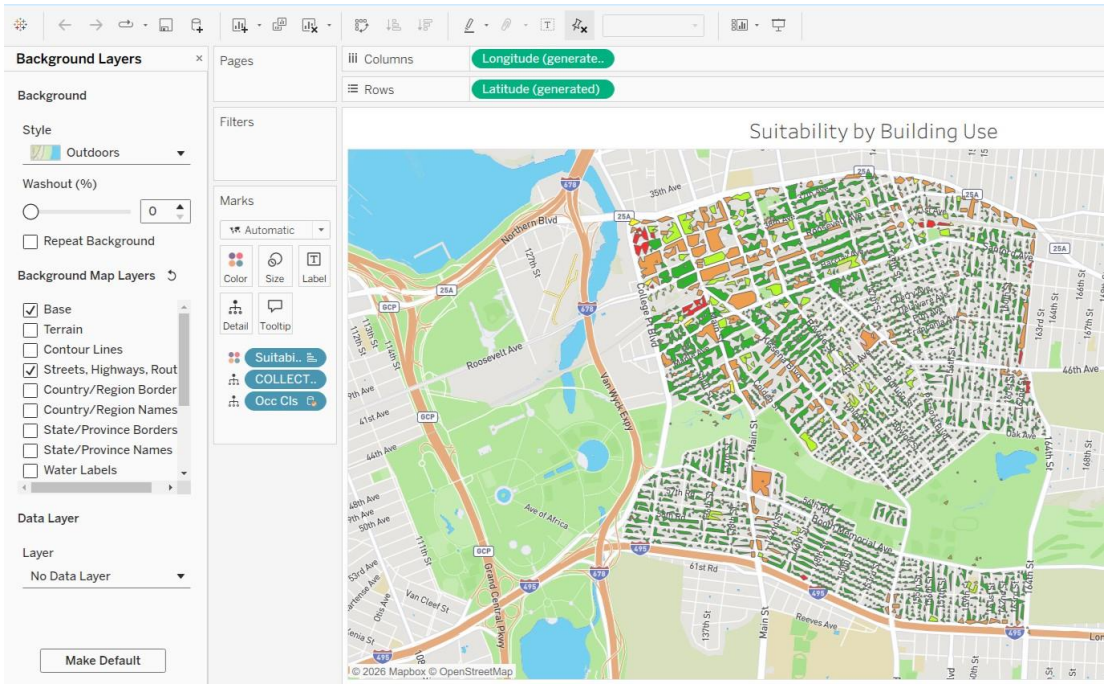

Figure 5. Tableau map showing "Suitability by Building Use" with the Outdoors background map style and Streets layer enabled.

6.3 Applying Consistent Colour Coding

To ensure visual consistency and comparability across all map sheets, a standardised colour palette was applied to each suitability layer. The colour scheme uses green for Highly Suitable, yellow-green for Suitable, yellow for Moderately Suitable, orange for Less Suitable, and red for Not Suitable — consistent with traffic-light-style convention familiar to planners and decision makers.

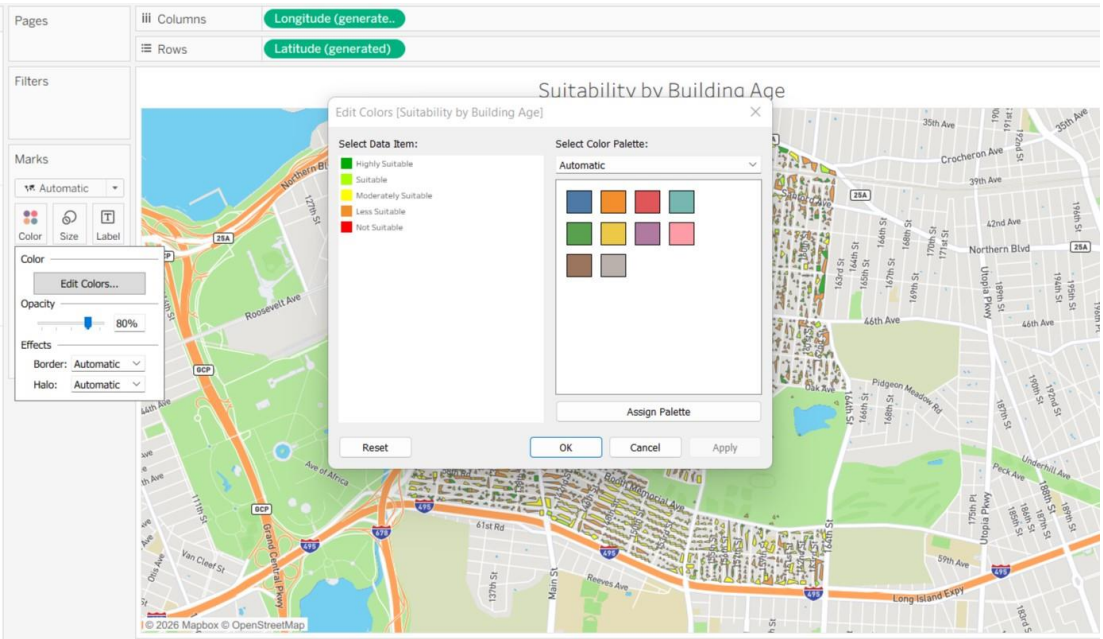

Figure 6. Tableau colour palette editor — assigning consistent colours to each suitability category for "Suitability by Building Age."

6.4 Configuring Tooltips

Responsive tooltips were enabled for all map sheets so that hovering over any building polygon displays its suitability classification immediately. The tooltip was configured to show the suitability field label, allowing end users to identify individual building classifications without additional interaction.

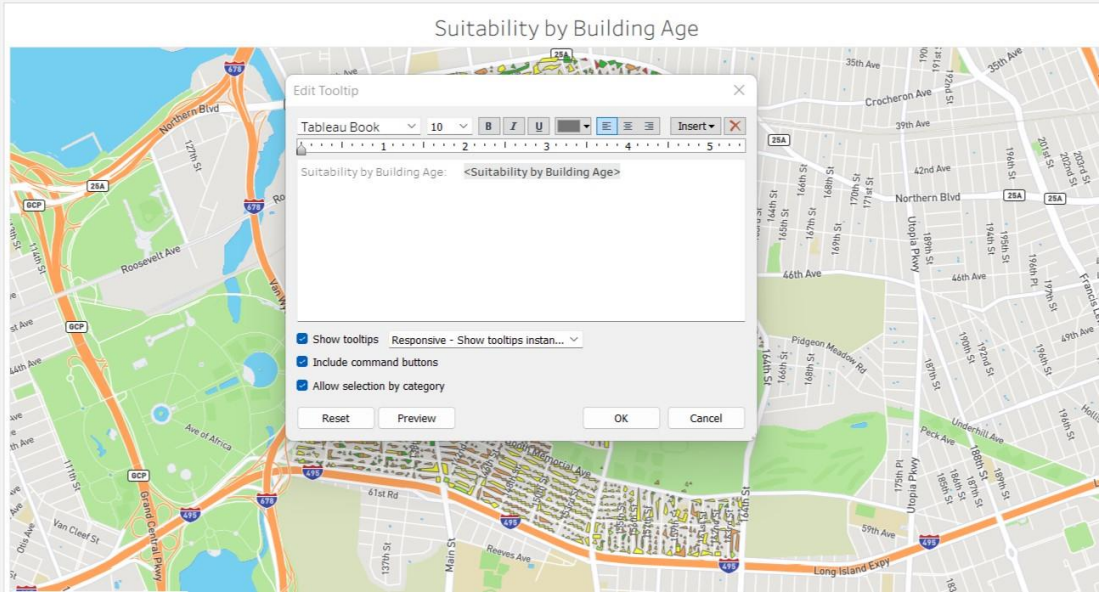

Figure 7. Tableau tooltip configuration for "Suitability by Building Age" — showing responsive tooltip with suitability classification field.

7. Step 5 — Dashboard Assembly and Tableau Public Publication

All individual criterion map sheets and the composite suitability map were assembled into a comprehensive Tableau dashboard. The same 1–5 scale and consistent colour coding across all sheets allows for straightforward comparison and supports the raster calculation logic. The dashboard was then published to Tableau Public for public access.

The published dashboard is accessible at:

`tableau.com/app/profile/john.antwi/viz/FinalExam_17784937562350/Dashboard1`

8. Step 6 — Embedding the Dashboard in the GitHub Repository

Following publication on Tableau Public, the embedding code was copied and refined for integration into the course GitHub Pages website. Claude.ai was used to refine the embed code to fit the minimal-skin Jekyll repository layout.

8.1 Obtaining and Refining the Embed Code

The Tableau Public embed code was copied directly from the published visualization. The HTML embed snippet was then refined — adjusting container widths, margins, and positioning — to render correctly within the Jekyll site's responsive layout.

```html
<div class="tableau-container" style="overflow-x: auto; margin: 2em 0;">
  <div class="tableauPlaceholder" id="viz1778558487202" style="position: relative;">
    <noscript>
      <a href="#">
        <img
          alt="Dashboard 1"
          src="https://public.tableau.com/static/images/Fi/FinalExam_17784937562350/
          style="border: none"
        />
      </a>
    </noscript>
    <object class="tableauViz" style="display: none;">
      <param name="host_url"             value="https%3A%2F%2Fpublic.tableau.com%2F"
      <param name="embed_code_version"   value="3" />
      <param name="site_root"            value="" />
      <param name="name"                 value="FinalExam_17784937562350/Dashboard1"
      <param name="tabs"                 value="no" />
      <param name="toolbar"              value="yes" />
      <param name="static_image"         value="https://public.tableau.com/static/im
      <param name="animate_transition"   value="yes" />
      <param name="display_static_image" value="yes" />
      <param name="display_spinner"      value="yes" />
      <param name="display_overlay"      value="yes" />
      <param name="display_count"        value="yes" />
      <param name="language"             value="en-US" />
      <param name="filter"               value="publish=yes" />
    </object>
  </div>
</div>

<script type="text/javascript">
  var divElement = document.getElementById("viz1778558487202");
  var vizElement  = divElement.getElementsByTagName("object")[0];

  if (divElement.offsetWidth > 800) {
    vizElement.style.width  = "1366px";
    vizElement.style.height = "3127px";
  } else if (divElement.offsetWidth > 500) {
    vizElement.style.width  = "1366px";
    vizElement.style.height = "3127px";
  } else {
```

Figure 8. Refined Tableau Public embed code (HTML) ready for insertion into the GitHub Pages repository markdown file.

8.2 Editing dashboards.md on GitHub

The refined embed code was pasted into the project's dashboards.md file within the docs/final-project/ directory of the GitHub repository. The markdown file was edited directly in the GitHub web interface to insert the HTML embed block.

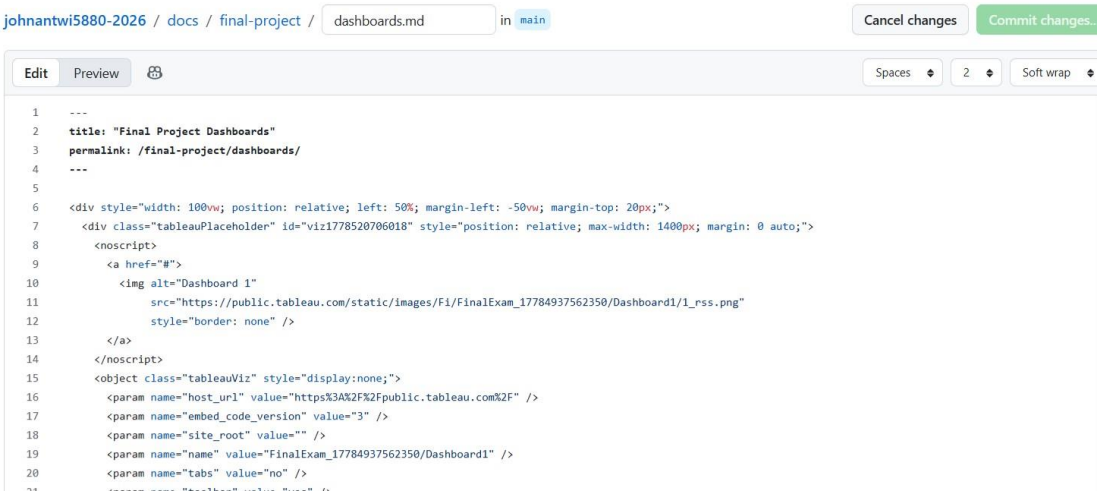

Figure 9. GitHub web editor showing the dashboards.md file with the Tableau embed code inserted, including the correct front matter title and permalink.

8.3 Verifying the GitHub Page Preview

After committing the changes, the rendered preview of dashboards.md was inspected on GitHub to confirm that the front matter was correctly structured and the page content was displaying as expected.

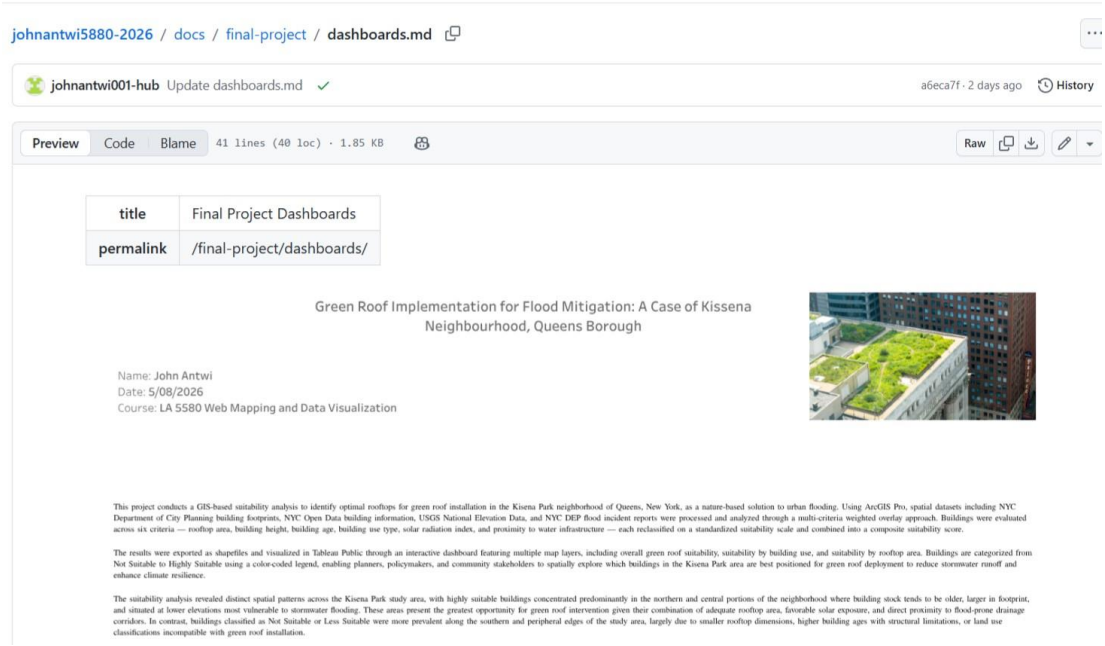

Figure 10. GitHub preview of dashboards.md showing the project title, author information, and introductory text for the Final Project Dashboards page.

8.4 Updating navigation.yml

To make the dashboard page accessible through the website navigation menu, the navigation.yml file in the docs/_data/ directory was updated. A new entry was added under the Final Project section with a title of "Final Project Dashboards" and the permalink /final-project/dashboards/, matching exactly the permalink declared in the dashboards.md front matter.

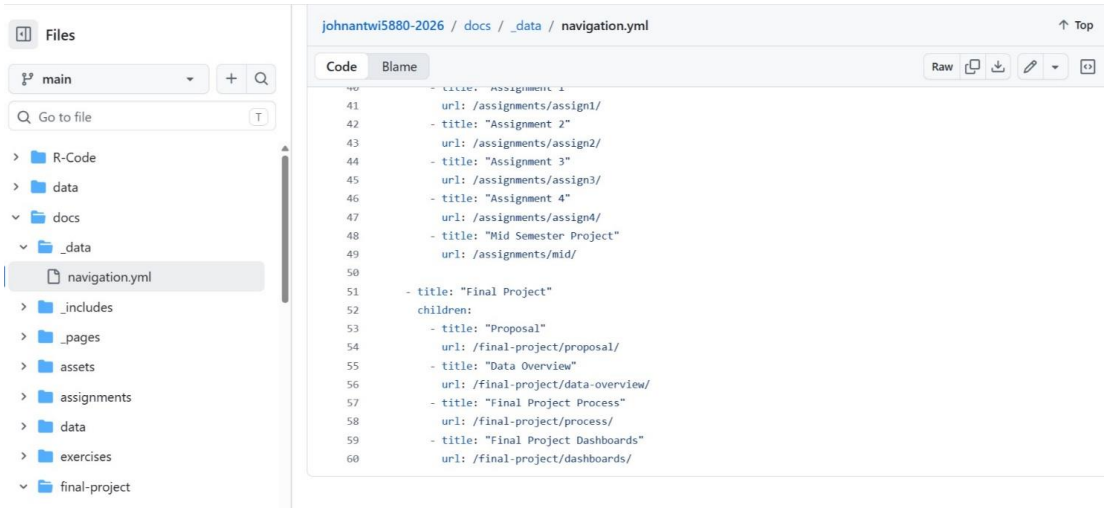

Figure 11. navigation.yml showing the Final Project section with matching permalink entries for Proposal, Data Overview, Final Project Process, and Final Project Dashboards pages.

9. Step 7 — Final Website Display Verification

After committing all changes and allowing GitHub Pages to rebuild the site, the final deployed website was verified. The interactive Tableau dashboard rendered correctly within the site's layout, with the navigation menu linking to the dashboard page as configured.

Figure 12. Final deployed GitHub Pages website showing the "Final Project Dashboards" page with the embedded Tableau dashboard for Green Roof Implementation for Flood Mitigation, Kissena Neighbourhood.

10. Summary

The project successfully delivered a comprehensive GIS-based suitability analysis for green roof implementation in the Kissena Neighbourhood of Queens Borough. The full workflow — from spatial data assembly in ArcGIS Pro through weighted overlay analysis, Tableau visualisation, and final publication on GitHub Pages — is documented in this report with supporting screenshots at each stage.

Key outputs include:

- A multi-criteria suitability assessment of all buildings in the Kissena Neighbourhood across six weighted criteria.
- Individual suitability maps for each criterion (rooftop area, sunlight, water availability, building age, building use, building height) visualised in Tableau with consistent colour coding and tooltips.
- A composite weighted overlay suitability map combining all six criteria.
- An interactive public Tableau dashboard published to Tableau Public.
- A fully functional GitHub Pages project website with embedded dashboard accessible via the course repository navigation.